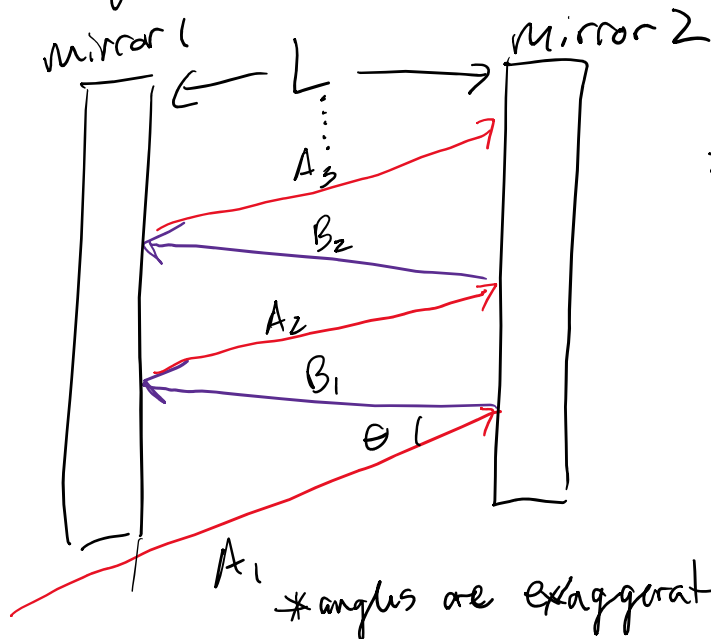


Fabry-Perot optical Resonant Cavity (etalons)



* mirror 1 and 2 have the same reflection coefficients

$|r|$ - reflection coefficient

$$K(2L); \quad k = \frac{2\pi}{\lambda}$$

↓

from A to B will be $2L$

* the light is coherent for all time and distance

* two beams A_1 and B_1 will interfere

$$A_1 + B_1 = A_1 + A_1 |r|^2 e^{-j(2kL)}$$

$B_1 = A_1 |r|^2$ because it is the initial intensity (irradiance)

* $A_1 = E_0$ that is affected by the interaction with the first mirror.

E-field in the cavity

$$\begin{aligned} E_{\text{cavity}} &= (A_1 + B_1) + (A_2 + B_2) + (A_3 + B_3) + \dots \\ &= A_1 + A_1 |r|^2 e^{-j(2kL)} + A_1 |r|^4 e^{-j(4kL)} + \dots \end{aligned}$$

Sum of Geometric Series

$$E_{\text{cavity}} \approx \frac{A_1}{1 - |r|^2 e^{-j(2kL)}} \rightarrow \text{after neglecting the higher order terms}$$

$$\begin{array}{ccc} \text{I} & \propto & |E|^2 \\ \uparrow & & \downarrow A_n \\ \text{intensity} & & \end{array}$$

$$\boxed{\text{I}_{\text{cavity}} = \frac{\text{I}_0}{(1 - \tilde{R})^2 + 4\tilde{R}\sin^2(kL)}}$$

I_{cavity} is maximum when $\sin^2(kL) = 0$

$kL = m\pi$; m is the Resonant mode (integer)
↙ looking at peaks of Brightness
 $m=6$



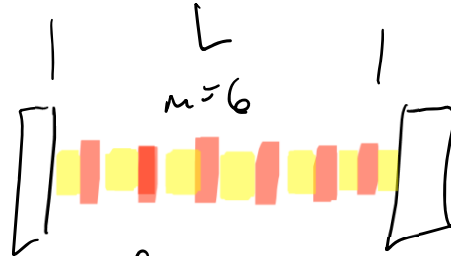
dark spots do not go completely to zero.

Spectral width (Δf_m): f_m is the mode freq. "Resonant freq"

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$$Finesse (F) = \frac{\pi \sqrt{R}}{1 - R}$$

$f_f \rightarrow$ fundamental freq.



$f_m \rightarrow$ mode 6 bright spots.

$$\Delta f_m = \frac{f_f}{F}$$

$\leftarrow$ contrast

b/w light and dark spots.

$$f_m = m f_f$$



$$m F = \frac{m f_f}{\Delta f_m} = \frac{f_m}{\Delta f_m}$$

$$m F = \frac{f_m}{\Delta f_m} \rightarrow \frac{f_m}{\Delta f_m} = \text{Quality factor (Q)}$$

$$m = \frac{2nL}{\lambda}$$

n is the index of refraction in the medium